

Senior Design 1 Final Report

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Executive summary

The objective of this project is to evaluate the technical feasibility of producing rocket propellant fuel from biomass, specifically woodchips, using a simulated process. Interest in biomass-derived propellants is driven by increasing demand for sustainable aerospace fuels, the availability of feedstocks, and favorable state incentives to support renewable fuel production. This study is intended to support early-stage process design and identify key performance drivers prior to experimental validation or detailed economic analysis.

The proposed process converts woodchip biomass into usable rocket propellant through a sequence of fast pyrolysis, bio-oil upgrading (hydrotreating), and fractionation. Process modeling is currently being conducted using ASPEN simulation software to represent major unit operations, mass and energy balance, cost analysis, utilities, and overall system behavior. At this stage, the simulation focuses on establishing a representative process flow diagram, defining reasonable operating assumptions, and identifying critical conversion steps that govern efficiency and yield. Detailed model calibration and optimization are ongoing.

Preliminary simulation results indicate that biomass-to-fuel conversion is achievable within realistic operating conditions. The model suggests that fuel production is strongly influenced by feedstock properties, conversion efficiency in the primary processing steps, and the overall energy integration within the system. While specific numerical results are still under development, insights into the key units that dominate material loss and energy demand have been obtained.

Based on the current findings of this project, biomass-derived rocket propellant production appears promising as a sustainable fuel pathway, especially in California's strict regulatory environment. However, further refinement of the ASPEN simulation and incorporation of economic evaluation are required to fully assess viability. Continued modeling efforts will focus

on improving confidence in predicted performance and on finding opportunities for process optimization.

1. Introduction

1.1 Problem & Significance

The space industry is expanding as many new space-related opportunities arise. The space industry is expected to hit \$1.8 trillion by 2035. In 2024, the US averaged 60 to 70 launches per week. The current RP-1 is made using fossil fuels, which have high Scope 1 emissions. Companies such as Amazon are adopting net-zero policies in their efforts to launch space satellites.

1.2 Supply & Demand Analysis

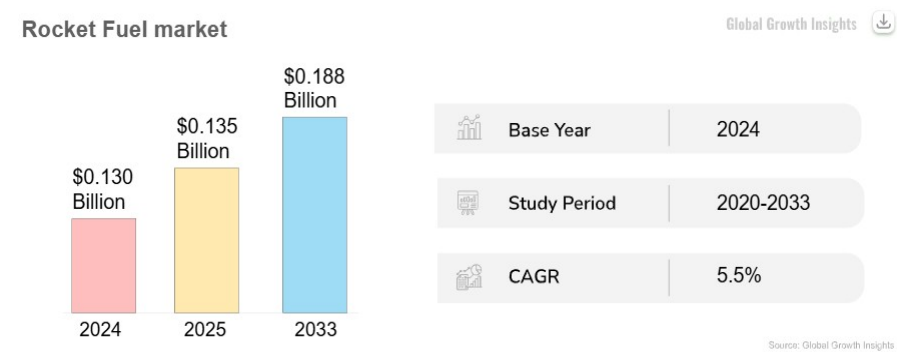


Figure 1 Projected market trend of the rocket fuel market

While there is no supply-and-demand curve for RP-1, sufficient information is available on the current market and its projected future direction. The figure above shows the current and projected values of the rocket fuel market. It is currently around \$0.130 billion and is projected to reach \$0.188 billion. This shows a CAGR of 5.5%.

The space industry is expected to hit \$1.8 trillion by 2035. It is currently at \$630 billion. This indicates an average annual growth rate of 9%. Since the vehicle-launch industry is a niche compared to other petroleum industries, there is a limited supply of RP-1 due to the few suppliers. The other industries that heavily rely on petroleum-based fuels include land vehicles and aviation. Since these industries are more popular and in higher demand than the vehicle launch industry, more focus is placed on them.

1.3 Global Share

Table 1 Location, global demand, and rocket fuel used

Location	Global Rocket Fuel Demand	Rocket Fuel Used
US	35-40%	LOX/RP-1, LOX/methane
Asia-Pacific	30-35%	LOX/LH2
Europe	20-25%	LOX/LH2
Middle East & Africa	5-10%	LOX/RP-1

The above table shows the location, global demand, and rocket fuel used. This table is significant since it shows that the US is a leader in demand and uses RP-1. This is helpful since the plant location is easily decided.

In 2024, the US averaged 60 to 70 launches per week. By comparison, China launched only 60 missions in 2023.

1.4 Value Chain Analysis

Table 2 Value chain analysis summary of chosen process

Stage	Value Added	Major Cost	Risk
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Biomass Supply	Process begins	Cost of biomass	Supply variability
Pyrolysis	Conversion of biomass into bio-oil	Reactor heat, sand recycle	Tar formation
Combustion Loop	Self-sustaining heat	Air compression	Sand attrition
Quench & Condensation	Captures vapors and converts to liquid	Cooling	Phase separation issues
Hydrotreating	Converts bio-oil to hydrocarbons	H ₂ , catalysts	Catalyst poisoning
Distillation	Separation	Reboiler energy	Coking
Hydrocracking	Produces fuel	H ₂ , pressure, furnaces	Over-cracking
Hydrogen Loop	Supplies H ₂	Natural gas	Reforming downtime
Final Fuel	Final product	Distribution	Fossil fuel competition

The table above illustrates the value chain analysis of our chosen process. This breaks down each stage, describing the value added, major costs, and risks associated with each stage. This is important since it allows for further investigation into improving the process and where major problems can arise.

1.5 Material Prices

Table 3 Required materials' prices

Continuous Feedstock	1 st Run	Continuous Product
Hybrid poplar wood chips	H2	RP-1
\$0.03/bone dry pound delivered	\$3/pound	\$4.60/pound

The above table shows the material prices. The continuous feedstock is hybrid poplar wood chips, which cost \$0.03 per bone-dry pound delivered. For the first run of our process, H2 is required, which costs \$ 3 per pound. Our continuous product is RP-1, which can potentially be sold for \$4.60/pound.

Sustainable RP-1 Price Calculation

- RP-1 from petroleum: \$1.04/lb – \$1.57/lb
- Sustainable aviation fuel (SAF) is around 2-5 times more expensive than fossil jet fuel
- Assuming the 2-5 times multiplier applies to rocket fuel
- Average RP-1 price: \$1.31/lb
- Average of the multiplier: 3.5
- Sustainable RP-1 price: \$4.60/lb

Since RP-1 is essentially a de-sulfurized kerosene, the assumption that the multiplier is similar is plausible. Since there is no information on the current price of sustainable RP-1, this is the best option.

2. Process Description

2.1 Competing Processes

Bio Propane

Bio-propane is a renewable form of propane produced through the hydrotreating of waste fats and bio-oil feedstocks. Its chemical properties are nearly indistinguishable from those of conventional propane, allowing it to be used in liquid-propane rocket engines with minimal modifications. Companies exploring bio-propane as a primary fuel include Orbex and BluShift Aerospace, driven largely by its sustainability benefits.

One of the key characteristics of bio-propane is its high energy density, which provides favorable mass efficiency for smaller launch systems. Its combustion is cleaner than that of kerosene-derived fuels, producing less soot and fewer sulfur oxides. Overall, its emissions profile is lower than its contemporaries. This results in extended engine component life and reduces the environmental impact of launches. Another advantage is its storage behavior: bio-propane can be stored as a liquid under moderate pressure, avoiding the storage complexities associated with cryogenic fuels.

However, bio propane carries several disadvantages. Production costs are high due to the limited availability of renewable propane and the hydrotreating processes required to generate it. These feedstock and processing limitations constrain the large-scale adoption of biofuels, keeping their prices significantly higher than those of petroleum-derived fuels. Additionally, although it offers cleaner combustion, propane engines can still experience

carbon deposition. These deposits increase maintenance needs and may offset some operational benefits.

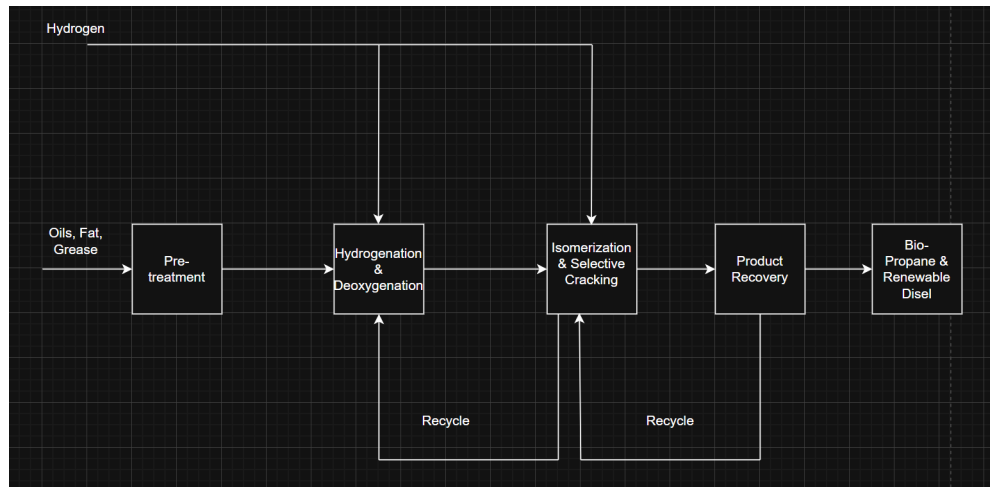


Figure 2: Propane Block Flow Diagram

Biomethane

Biomethane is a renewable, chemically identical substitute for fossil methane that can be produced through anaerobic digestion, biomass gasification, or synthetic-gas methanation. It has become an increasingly attractive rocket-fuel candidate as the industry explores high-performance methalox propulsion systems. Companies pursuing this include Interstellar Technologies and Air Water Hokkaido Inc., a collaboration between a space infrastructure company and an industrial gas producer.

One of biomethane's major advantages is that its production is commercially established on a global scale. Biogas upgrading to biomethane is common in the energy sector, and this

maturity translates to stable supply chains. It offers cleaner combustion than kerosene, producing less particulate matter and reducing greenhouse gas impact when its renewable origin is considered. From a performance standpoint, methane-fueled rocket engines deliver high specific impulse and improved thermal efficiency, making them attractive for reusable launch systems and future deep-space missions.

Despite these advantages, biomethane presents significant engineering and logistical challenges. Methane engines are inherently more complex than kerosene engines, requiring precise cooling strategies, advanced turbomachinery, and tightly controlled combustion conditions. The fuel must also be stored cryogenically at low temperatures, necessitating insulated tanks, boil-off management systems, and additional ground infrastructure. Furthermore, producing biomethane from biomass often requires multiple upgrading steps, adding cost and technical complexity to the process.

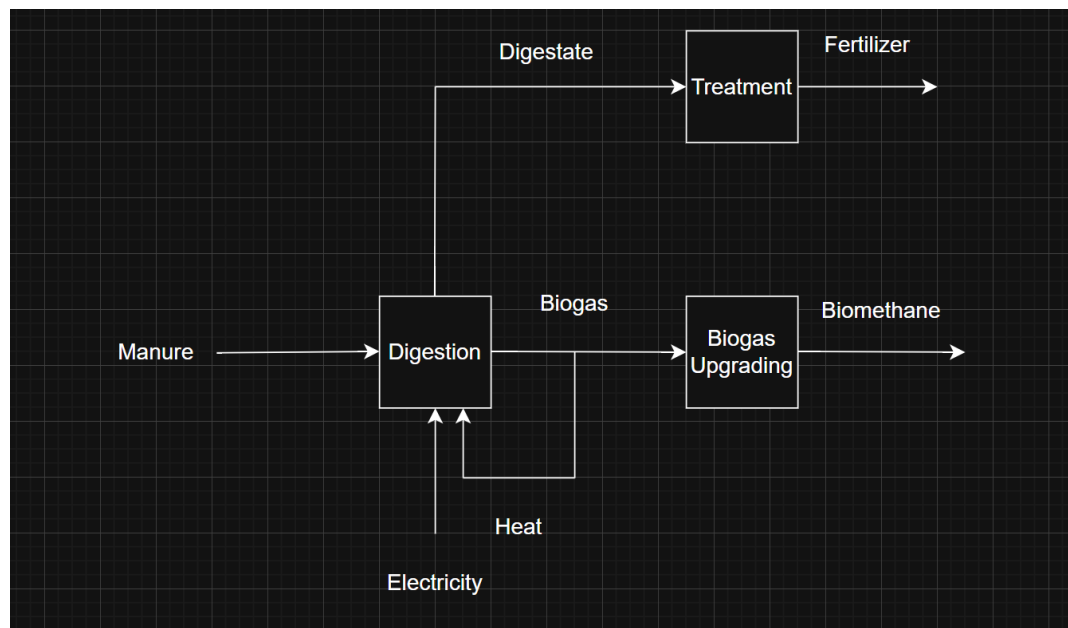


Figure 3: Methane Block Flow Diagram

Standard Crude Oil

Crude-oil-derived rocket fuels, such as RP-1, remain the aerospace industry's most widely used hydrocarbon propellants. The fuel is produced through petroleum-refining processes that have been optimized over decades. Major users include SpaceX, Northrop Grumman, and the United Launch Alliance (ULA). The global refinery infrastructure supporting these fuels is mature, enabling a consistent supply at large volumes and at relatively low cost.

As mentioned previously, the largest benefit of crude-oil-derived fuels is their low production expense. The worldwide oil industry is already equipped to produce kerosene at a massive scale; as such, RP-1 production benefits from economies of scale not available to biofuels. Crude-derived fuels are easily stored at ambient temperatures and do not require cryogenic systems, thereby reducing complexity in tank design and launch-pad logistics. Additionally, the refinery processes used to produce RP-1 do not require the specialized upgrading steps typically associated with renewable-fuel production, making petroleum fuels simpler and more economical to manufacture.

However, there are several drawbacks, particularly in terms of long-term sustainability and environmental impact. Crude-derived fuels produce some of the highest carbon emissions among aerospace propellants, and their combustion generates soot and black carbon that can accumulate in the upper atmosphere. As global launch rates continue to rise, these environmental impacts cannot be ignored. Crude oil is also a finite resource; its deposits are limited within the Earth's crust. Additionally, producing RP-1 still requires deep

desulfurization and extensive purification to remove contaminants such as sulfur, nickel, and vanadium. These processes increase energy demand and contribute to its environmental footprint further.

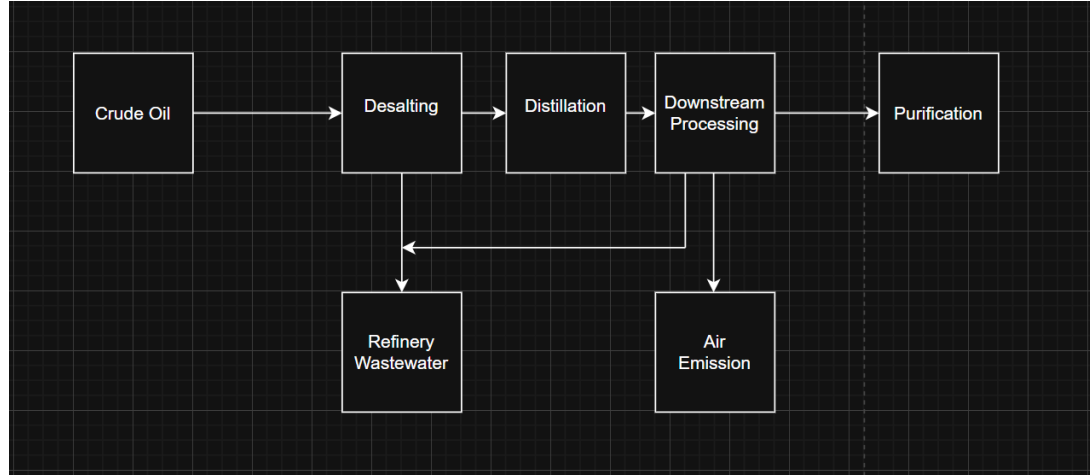


Figure 4: Standard Crude Oil Block Flow Diagram

2.2 Chosen Process

The chosen process of going from biomass to bio-oil has its complexities and constraints in the production process. The first stage of the process involves creating bio-oil from wood chips via fast pyrolysis. The process begins with the biomass entering a dryer to ensure it has zero to no moisture before entering the pyrolysis riser reactor, which operates at a nominal temperature of 500 °C to 930 °C. The effluent exits from the top, entering a cyclone that separates solids from the moisture. The moisture is carried up into stream 103 to undergo heat exchange and then heated to be quenched in the quench absorber. Following the outlet stream 106, the bio-oil then gets pressurized and heated before further processing. From the other outlet of the quench absorber stream 111, the vapor is compressed and then heat-exchanged to be fed as a fluidizing gas for the pyrolysis reactor. Returning to the cyclone, stream 115, a stream of solid, is fed into a combustor reactor at a temperature

between 500 °C and 1000 °C, resulting in wet ash. From the wet ash stream, 116 gets sent to a cyclone to separate moisture from ash. The ash in stream 117 is still moist at a relatively low level. Stream 118, which is the moisture stream, gets heated up and exchanged with air. The air is used to feed the char combustor. The stream 124 then becomes flue gas, which heats the dryer and is exhausted through the exhaust stream.

Once bio-oil is produced, it must undergo hydrotreating to stabilize and upgrade the oil for use as a fuel. This is due to the nature of bio-oil being highly saturated with ~25% water, making it difficult to separate. Thus, it is unstable for long-term storage and not miscible in any conventional hydrocarbon-based fuels. The incoming bio-oil stream 200 is pressurized into a stream of H₂, which is then heat-exchanged into the first hydrotreater, R-200, running at 240 °C and 2500 psig. The effluent from the hydrotreater, stream 221, is then heat-exchanged and heated in a furnace to approximately 300 °C. The heated stream is then fed into the second hydrotreater, which operates at 370 °C and 2015 psig. The reactor effluent is then heat-exchanged twice with an air cooler and further cooled in a heat exchanger using cooling water. Stream 207 is then fed into a high-pressure flash, which separates the gas from the oil. Following the liquid stream 208, it is then low-pressure flashed and heated in an exchanger in the bottom oil product. The overhead stream 213, which is then letting off fuel, will be fed into reformers. Following stream 215, the vapor stream is then fed into a pressure swing adsorption unit (PSA) to recover some H₂ gas. Following the PSA stream, 217 is then fed a makeup H₂ stream and subsequently compressed before being returned to the hydrotreaters.

The final stage of this design involves converting stable oil to RP-1. Once the oil from the bottoms is created, the oil is then fed to a debutanizer distillation. Where the distillate is fuel gas to be fed to heaters and reformers. The bottom product is then the stable oil that will be fed into a hydrodesulfurizer, R-300. The reactor effluent is then fed into a flash drum to separate the gas from the stable oil. The tops, which are stream 307, will yield fuel gas and H₂ for recycling in the hydrodesulfurizer. The bottom, which is a stable oil, is then fed into a furnace to be heated before undergoing fractional distillation. From fractional distillation, a hard cut from the middle of the column is taken just before the dew point for the RP-1 kerosene (C₁₂-C₁₆). The distillate yields relatively low octane naptha gas that can be sold as is or as a blend. The bottom product yields completely renewable and usable diesel that can also be sold.

3. Appendix

3.1 Design Basis

The plant capacity was selected as 280 million pounds per year of RP-1, based on production rates reported in an existing, published process described in the article cited below.[6]

Three primary criteria guided the selection of the plant location: feedstock availability, tax and policy incentives, and proximity to major aerospace and launch facilities. California meets all three criteria exceptionally well.

California is one of the largest lumber-producing states in the U.S., providing a large and reliable supply of wood-based feedstocks, such as wood chips, which are produced directly or indirectly from lumber processing operations. This ensures long-term feedstock availability at a lower rate. [7]

In addition, California offers significant financial incentives for low-carbon and renewable fuel production, including programs that improve the projects' profitability through tax benefits and carbon-reduction credits.[11]

Finally, California is home to major aerospace and launch organizations, including SpaceX, NASA, and Northrop Grumman, which operate launch sites and research centers within the state. Proximity to these organizations supports easier fuel distribution, potential partnerships, and long-term demand for our renewable RP-1.

3.2 Block Flow Diagram (BFD)

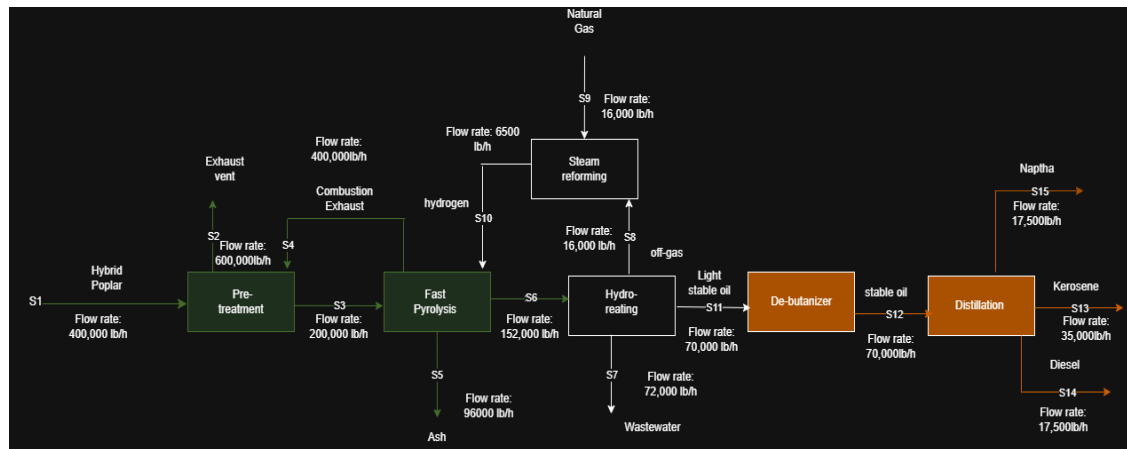


Figure 5: Block Flow Diagram

3.3 Preliminary Process Flow Diagram



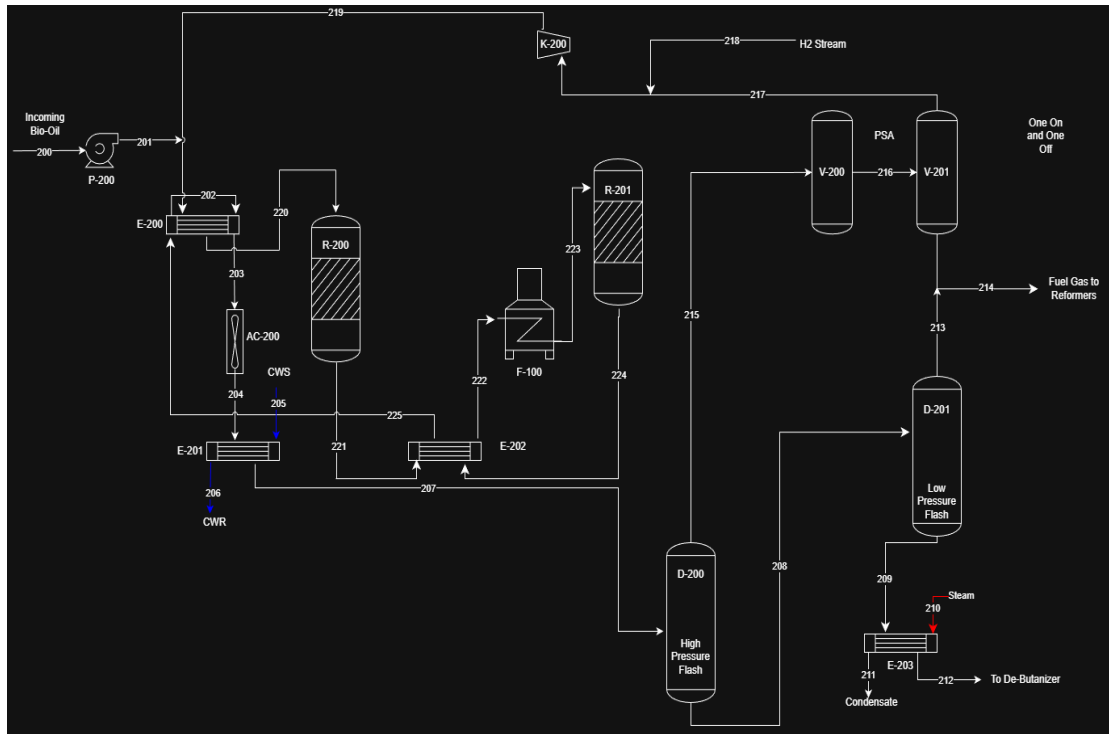


Figure 7:PFD for Bio oil upgrading

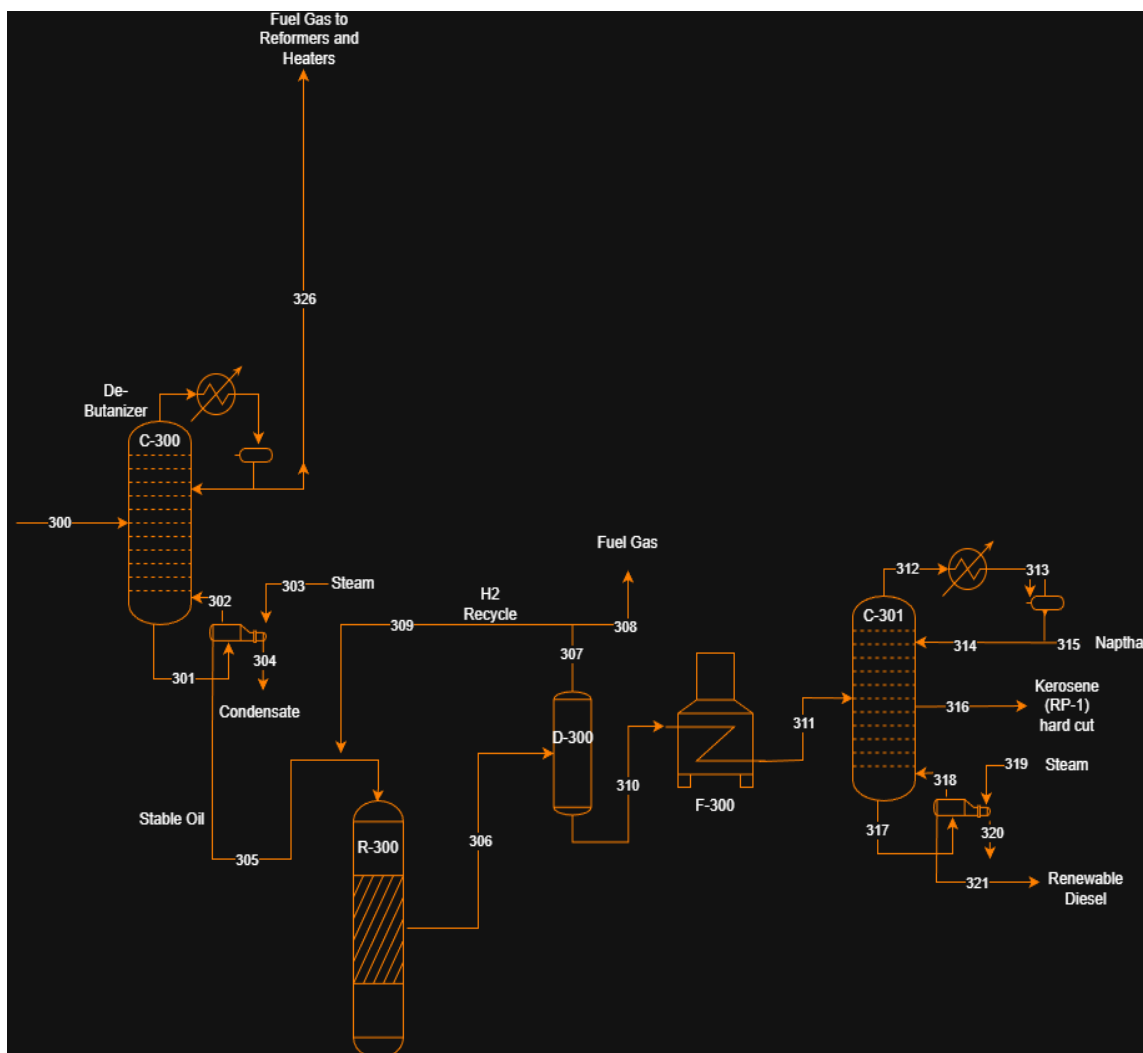


Figure 8: PFD RP1 separation

3.4 Rough Material Balance

Table 4: General Material Balance around the BFD

Block	Pre-treatment	Fast Pyrolysis	Hydro-treating	De-butanizer	Distillation	Steam reforming
Stream	$S1+S4=S2+S_3$	$S3+S10=S4+S5$	$S6=S7+S11+S8$	$S11=S12$	$S12=S13+S14+S15$	$S8+S9=S10$

specified variable is the mole fraction of O_2 at 0.02 at S13 for Aspen to iteratively adjust its input Air flow to meet its desired amount directly for perfect execution and matching perfectly its desired amount of oxygen exiting the combustor at 2%. This amount is 2% because it is expected to behave similarly to industrial requirements for excess air requirements to guarantee complete burning of char while avoiding excess dilution from nitrogen addition to flue gas for industrial applications properly, while maintaining its heat correctly for repeated reheating for its primary role at the pyrolysis reactor to guarantee successful execution of its processes properly.

A further specification was added to the Aspen Plus input file to ensure adiabatic operation of the RYIELD reactor (R-100), specifying a reactor duty of 0 kW, which means no heat addition or removal is to occur. In this specification, the reactor duty for R-100 was specified as the specified variable and maintained at its target value of 0 kW, while the reactor temperature for E-103 was specified as the manipulated variable that Aspen should adjust to meet this condition. It should operate within reactor temperatures ranging from 100°C to 800°C, ensuring this is a realistic range for which Aspen can determine the correct inlet temperature to self-heat to its final condition solely by reaction heat, without requiring additional heat input from external heating sources. This is also realistic for fast pyrolysis reactor designs, where biomass is heated solely by hot-sand circulation rather than requiring additional heat input components.

It is logical to use reactor temperature as this is directly dependent on whether or not reactor energy terms will reach closure without requiring heat interaction rather than being dependent on E-103 manipulated variable at all points during simulates reactor processes using Aspen plus software because inlet temperature to reactor-R100 is one factor that mainly affects whether or not energy terms reach closure at all points during simulates reactor processes using Aspen plus software because inlet temperature to reactor-R100 is one factor that mainly affects whether or not energy terms reach closure at all points during reactor processes.

The pyrolysis reactor was modeled using wood as the primary biomass feedstock. Because wood is a nonconventional solid, its physical-chemical behavior in Aspen Plus was defined using proximate analysis, ultimate analysis, and sulfur analysis. These values allow Aspen's nonconventional solid property models (HCOALGEN and DCOALIGT) to calculate sensible enthalpy, density, and heating values. The assignment indicated that the biomass contained 10% moisture, so the raw wood stream was dried prior to pyrolysis. This drying step removes only water, ensuring the reactor receives fully dried wood. The same proximate, ultimate, and sulfur analyses defined for wood were also used as the basis for downstream modeling of ash, with char receiving updated values reflecting its expected high fixed carbon, low volatile matter, and greatly reduced oxygen and hydrogen after pyrolysis.

The pyrolysis reactor (R-100) was modeled as an RYield reactor. The RYield block was used in this assignment since the detailed kinetics and reaction pathways were not provided, and the desired behavior was instead defined by experimentally measured pyrolysis reactor yields. In this model, each product stream from the assignment was assigned a mass-based yield for every 100

grams of dry wood entering the reactor. These yields included hydrogen, carbon monoxide, carbon dioxide, water, methane, ethylene, levoglucosan, guaiacol, acetone, acetic acid, cresol, furfural, and char. Sand was listed as an inert component so that it passes through the reactor unchanged. Aspen automatically normalizes these yields to maintain mass balance. Because the pyrolysis process is strongly endothermic, the internal heat balance of the reactor must be satisfied by the sand loop supplying heat.

To enforce adiabatic operation, a design specification (DS-ADIAB) was created. The specification forces the calculated reactor duty to be zero by manipulating the temperature of the incoming hot sand. The manipulated variable was the temperature of the heater block feeding hot sand into the reactor, and it was allowed to vary within physically reasonable bounds. During simulation, Aspen adjusts the sand temperature iteratively until the reactor requires no external heat or cooling, meaning that the sand supplies exactly the amount of heat needed to achieve the specified yield distribution. This approach mirrors the behavior of circulating fluidized-bed pyrolysis systems, where heat is transferred from a recirculating solid heat carrier rather than from an external furnace or internal heating elements.

Several assumptions were necessary to implement this model. Wood is treated as a uniform material with fixed proximate and ultimate analysis values, even though natural biomass varies significantly. The RYield reactor assumes that the product slate depends only on the specified yields; no reaction kinetics, residence-time effects, or secondary cracking reactions are considered. Gas-phase secondary reactions such as tar cracking or polymerization were neglected. Sand and ash were treated as chemically inert solids. The reactor was assumed to be perfectly mixed and adiabatic, with all heat provided solely by the incoming sand. Vapor-phase components were assumed to behave ideally under the selected property method. Sulfur was assumed to remain primarily in ash or char unless explicitly represented in a vapor-phase product. Finally, the model assumes stable solids circulation and no agglomeration or defluidization behavior.

The char combustor was modeled using a two-stage RStoic configuration to represent the sequential oxidation and burnout of char produced in the pyrolysis reactor. The goal was to recover thermal energy from char combustion, generate a high-temperature fluidizing medium for the pyrolysis reactor, and produce a clean flue gas stream suitable for downstream heat recovery. The combustor was therefore modeled as a combination of R-200A (char oxidation stage) and R-200B (gas burnout and flue gas polishing stage), each operating at controlled pressure and near-zero duty to achieve a thermodynamically consistent representation of adiabatic combustion.

The first reactor, R-200A, receives the char stream generated in fast pyrolysis (DRYCHAR) and initiates the decomposition of the nonconventional char component into its elemental constituents. Because char is defined in Aspen using nonconventional component modeling (with PROXANAL, ULTANAL, and SULFANAL attributes), the combustion chemistry must be

explicitly defined through stoichiometric reactions. In R-200A, we modeled the breakdown of char into elemental carbon (solid C), oxygen (O₂), nitrogen (N₂), hydrogen (H₂), sulfur (S), and inert ash. This step does not represent the physical chemistry of combustion itself, but rather converts the NC component into conventional components so Aspen Plus can compute stoichiometric oxidation in subsequent units. To achieve this, R-200A was configured with a fractional conversion of 1.0 on char, and with stoichiometric coefficients calculated using a custom calculator. Since all ash present in the char should remain solid throughout the combustion train, the PROXANAL and ULTANAL values for ASH were set to 100% ash and 0% of all other constituents to ensure the material balances were preserved.

Following char decomposition, the now-conventional elemental components flow to R-200B, which performs the true combustion calculations. R-200B was configured at 1.12 atm and set to “Generate Combustion Reactions” in Aspen Plus, which automatically constructs oxidation pathways for carbon, hydrogen, sulfur, and any remaining combustibles. Nitrous oxide (NO) was selected as the preferred nitrogen oxide product to reflect typical fluidized bed combustion conditions. In this reactor, all available oxygen from the air inlet is consumed stoichiometrically, producing CO₂, H₂O, SO₂, and trace amounts of NO. R-200B operates with a duty fixed at 0, making the reaction adiabatic. This is critical to representing the direct transfer of combustion heat to the circulating bed material without artificially imposing external heat addition or removal.

The thermal energy generated in R-200B is carried by the high-temperature flue gas and by the heated circulating solids stream (ASHSAND). These solids consist of the recycled bed material that loops back to the pyrolysis reactor to supply process heat. Cyclone (CYC-102) was used to mechanically separate coarse ash and fines from the flue gas, ensuring that unburned solids remain within the circulating bed loop while fine particulates exit via ASHFINES. The flue gas exiting R-200B and associated separators is sent to a series of heat exchangers (E-101, E-102, and E-100, depending on loops) to recover energy before being discharged as F-LUEGAS1 or F-LUEGAS2.

Air addition to R-200B was modeled explicitly via a dedicated air stream at 25°C and 1 atm. The amount of air supplied exceeds the stoichiometric requirement to ensure complete burnout of char and oxidation of volatile elements. Compressors and heat exchangers upstream of R-200B are included where needed to reflect required operating conditions but do not alter the fundamental combustion chemistry. The pressure was maintained at 1.12 atm throughout the combustor system to approximate slightly pressurized fluidized bed operation and to maintain consistency with the pyrolysis reactor.

Assumptions were required to construct a workable and thermodynamically consistent combustor model. It was assumed that char undergoes complete conversion in R-200A, with no

residual volatile matter remaining after pyrolysis. The ULTANAL elemental distribution was assumed constant for all char fed to the combustor, even though real char composition varies with heating rate, feedstock, and residence time. All ash was treated as inert and nonreactive, with no catalytic or heat capacity effects beyond those captured by Aspen's NC property models. Combustion was assumed to be adiabatic, with no heat losses to the environment, and all reactions in R-200B were treated as kinetically unlimited and instantaneous. Air was supplied at standard composition, and NO formation was limited to thermodynamic product selection rather than detailed NO_x kinetics. Finally, bed material inventory was assumed constant, and attrition losses were represented only through ASHFINES removal.

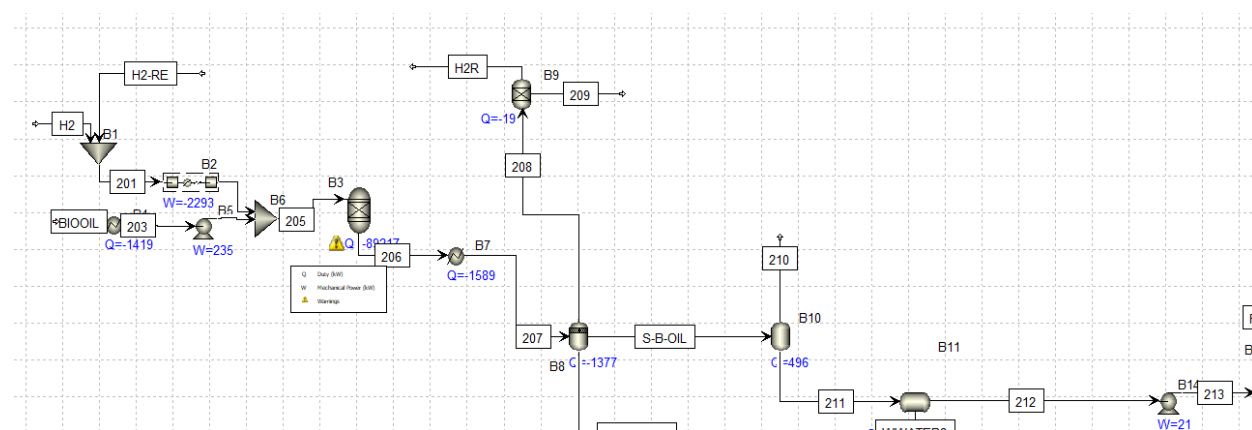


Figure 3: ASPEN PFD UPGRADING BIO-OIL

The filtered bio-oil product from the pyrolysis unit is pumped to high pressure, then combined with compressed hydrogen and preheated with reactor effluent. Two catalytic reaction stages are used. The first-stage catalytic reactor serves to stabilize the bio-oil through mild hydrotreatment over a cobalt-molybdenum (CoMo) hydrotreating catalyst. The product oil is further processed in the second-stage hydrotreater. The second-stage hydrotreater operates at a higher temperature and a lower space velocity than the first-stage hydrotreater. The CoMo catalyst is also used in this reactor. Reactor effluent and a fired heater provide feed preheat for the second stage. The second-stage product is separated into product oil, wastewater, and off gas streams.

The off-gas from the hydrotreaters is sent to a Pressure Swing Adsorption (PSA) system for recovery of the hydrogen gas. The recovered hydrogen is recycled back to the reactors. The low-pressure PSA tail gas stream, which is rich in byproduct light hydrocarbons, is sent to the steam reformer for hydrogen production. A small portion of this off-gas is used in the fired heater to preheat the feed to the second-stage hydrotreater.

Anaerobic digestion can be used to treat wastewater with an organic content of less than 2%.

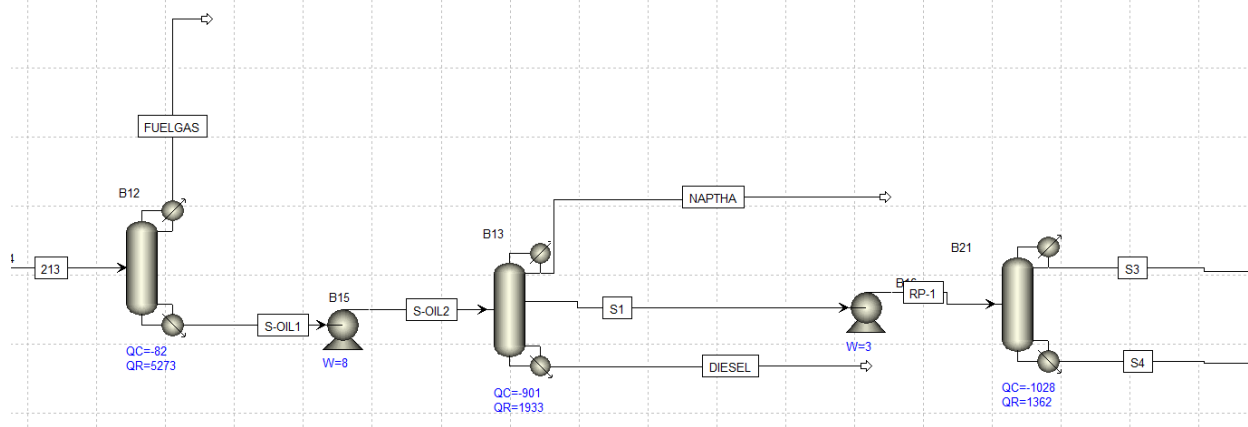


Figure 4: ASPEN PFD FRACTIONATION

This PFD shows the flow diagram for the hydrocracking and product separations areas. The hydrotreated oil is stabilized by removing the butane and lighter components in a debutanizer column. The stable oil stream is then separated into light and heavy fractions. The desired product, RP-1, which is the carbon range C9-C16, is extracted from a hard cut down in the middle. Side products such as Diesel (C12-C20) and Naptha (C5-C7) are also obtained from the tops and bottoms.

3.6 MASS BALANCE (Fast Pyrolysis)

Column1	Column2	BIOMASS (100)	HOTSAND (123)	FLUIDGAS (113)
Temperature	F	77	1129.9	707
Pressure	psi	33.8	33.8	33.94
H2 lb/hr	lb/hr	0	0	0
N2 lb/hr	lb/hr	0	0	0
CO lb/hr	lb/hr	0	0	17605
O2 lb/hr	lb/hr	0	0	0
CH4 lb/hr	lb/hr	0	0	847
C2H4 lb/hr	lb/hr	0	0	853.7
CO2 lb/hr	lb/hr	0	0	17835.5
SO2 lb/hr	lb/hr	0	0	1174.9
NO2 lb/hr	lb/hr	0	0	278.1

C3H6O-1 lb/hr	lb/hr	0	0	33088.4
H2O lb/hr	lb/hr	0	0	5478.4
C2H4O2-1 lb/hr	lb/hr	0	0	0
C5H4O2 lb/hr	lb/hr	0	0	0
C7H8O-4 lb/hr	lb/hr	0	0	0
C7H8O2-E1 lb/hr	lb/hr	0	0	0
C6H10O5-N1 lb/hr	lb/hr	0	0	0
SAND lb/hr	lb/hr	0	771617.9	0
ASH lb/hr	lb/hr	0	0	0
WOOD lb/hr	lb/hr	330693.4	0	0
CHAR lb/hr	lb/hr	0	0	0

Column1	Column 2	200	300	301	302
Temperature	F	896	896	2157.1	2157.1
Pressure	psi	29.4	29.4	16.5	16.4
H2 lb/hr	lb/hr	297.6	0	0	0
N2 lb/hr	lb/hr	0	0	0	5
CO lb/hr	lb/hr	34927.6	0	0	0
O2 lb/hr	lb/hr	0	0	0	17118.9
CH4 lb/hr	lb/hr	2424.4	0	0	0
C2H4 lb/hr	lb/hr	2103.7	0	0	0
CO2 lb/hr	lb/hr	35782.2	0	0	187625.8
SO2 lb/hr	lb/hr	1174.9	0	0	659.9
NO2 lb/hr	lb/hr	278.1	0	0	145.3
C3H6O-1 lb/hr	lb/hr	70886.6	0	0	0
H2O lb/hr	lb/hr	82596	0	0	63634.6
C2H4O2-1 lb/hr	lb/hr	33036.2	0	0	0
C5H4O2 lb/hr	lb/hr	8631.1	0	0	0
C7H8O-4 lb/hr	lb/hr	11904.9	0	0	0
C7H8O2-E1 lb/hr	lb/hr	81846.6	0	0	0
C6H10O5-N1	lb/hr	13095.5	0	0	0

lb/hr					
SAND lb/hr	lb/hr	771617.9	771617.9	771617.9	0
ASH lb/hr	lb/hr	0	0	0	86.6
WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	28869.5	28869.5	0	0

Column1	Column2	303	304	305	400
Temperature	F	2157.6	482	212	896
Pressure	psi	16.4	16.4	14.6	29.3
H2 lb/hr	lb/hr	0	0	0	297.6
N2 lb/hr	lb/hr	513905.2	513905.2	513876.3	0
CO lb/hr	lb/hr	0	0	0	34927.5
O2 lb/hr	lb/hr	17118.9	17118.9	156033.3	0
CH4 lb/hr	lb/hr	0	0	0	2424.4
C2H4 lb/hr	lb/hr	0	0	0	2103.6
CO2 lb/hr	lb/hr	187625.7	187625.7	0	35782.1
SO2 lb/hr	lb/hr	659.8	659.8	0	1174.8
NO2 lb/hr	lb/hr	145.2	145.2	0	278.1
C3H6O-1 lb/hr	lb/hr	0	0	0	70886.6
H2O lb/hr	lb/hr	63634.6	63634.6	0	82596.1
C2H4O2-1 lb/hr	lb/hr	0	0	0	33036.3
C5H4O2 lb/hr	lb/hr	0	0	0	8631.1
C7H8O-4 lb/hr	lb/hr	0	0	0	11904.9
C7H8O2-E1 lb/hr	lb/hr	0	0	0	81846.6
C6H10O5-N1 lb/hr	lb/hr	0	0	0	13095.5
SAND lb/hr	lb/hr	0	0	0	0
ASH lb/hr	lb/hr	0	0	0	0
WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	0	0	0	0

Column1	Column2	401	402	403	404
Temperature	F	807.8	113	113	113
Pressure	psi	29.4	24.9	24.9	24.9
H2 lb/hr	lb/hr	297.6	297.6	0	297.6
N2 lb/hr	lb/hr	0	0	0	0
CO lb/hr	lb/hr	34927.6	34927.6	0	34927.6

O2 lb/hr	lb/hr	0	0	0	0
CH4 lb/hr	lb/hr	2424.4	2424.4	0	2424.4
C2H4 lb/hr	lb/hr	2103.6	2103.7	0	2103.7
CO2 lb/hr	lb/hr	35782.1	35782.1	0	35782.2
SO2 lb/hr	lb/hr	1174.8	1174.8	0	1174.9
NO2 lb/hr	lb/hr	278.1	278.1	0	278.1
C3H6O-1 lb/hr	lb/hr	70886.6	70886.6	2835.5	68051.2
H2O lb/hr	lb/hr	82596.1	82596.1	66076.9	16519.2
C2H4O2-1 lb/hr	lb/hr	33036.3	33036.7	33036.7	0
C5H4O2 lb/hr	lb/hr	8631.1	8631.1	8631.1	0
C7H8O-4 lb/hr	lb/hr	11904.9	11904.9	11904.9	0
C7H8O2-E1 lb/hr	lb/hr	81846.6	81846.6	81846.6	0
C6H10O5-N1 lb/hr	lb/hr	13095.5	13095.5	13095.5	0
SAND lb/hr	lb/hr	0	0	0	0
ASH lb/hr	lb/hr	0	0	0	0
WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	0	0	0	0

Column1	Column2	409	410	BIO-OIL	AQUEOS
Temperature	F	176	707	191.6	191.6
Pressure	psi	33.9	33.9	24.9	24.9
H2 lb/hr	lb/hr	142.1	142.1	0	0
N2 lb/hr	lb/hr	0	0	0	0
CO lb/hr	lb/hr	16681.7	16681.6	0	0
O2 lb/hr	lb/hr	0	0	0	0
CH4 lb/hr	lb/hr	1157.9	1157.93	0	0
C2H4 lb/hr	lb/hr	1004.7	1004.73	0	0
CO2 lb/hr	lb/hr	17089.8	17089.8	0	0
SO2 lb/hr	lb/hr	561.1	561.1	0	0
NO2 lb/hr	lb/hr	132.8	132.8	0	0
C3H6O-1 lb/hr	lb/hr	32501.7	32501.7	2834.2	1.17
H2O lb/hr	lb/hr	7889.7	7889.7	67.6	66009.2
C2H4O2-1 lb/hr	lb/hr	0	0	32789.2	246.9
C5H4O2 lb/hr	lb/hr	0	0	8598.7	32.4
C7H8O-4 lb/hr	lb/hr	0	0	11902.9	2.00
C7H8O2-E1 lb/hr	lb/hr	0	0	81792.3	54.2

lb/hr					
C6H10O5-N1					
lb/hr	lb/hr	0	0	13004.8	90.6
SAND lb/hr	lb/hr	0	0	0	0
ASH lb/hr	lb/hr	0	0	0	0
WOOD lb/hr	lb/hr	0	0	0	0
CHAR lb/hr	lb/hr	0	0	0	0

3.7 MASS BALANCE (BIO-OIL UPGRADE to RP-1)

Column1	Column2	H2	BIOIL	H2-RE
Temperature	F	77	191.6	149
Pressure	psi	300	24.9	414.6
H2	lb/hr	6500	0	3041.9
CH4	lb/hr	0	0	0
C2H4	lb/hr	0	0	0
CO2	lb/hr	0	0	0
ACETONE	lb/hr	0	2834.2	0
H2O	lb/hr	0	67.6	0
AACID	lb/hr	0	32789.2	0
FURFURAL	lb/hr	0	8598.7	0
MCRESOL	lb/hr	0	11902.9	0
GUAIACOL	lb/hr	0	81792.3	0
LVG	lb/hr	0	13004.8	1.36761E-09
C2H6	lb/hr	0	0	0
C10H22	lb/hr	0	0	0
2-PHE-01	lb/hr	0	0	0
N-HEP-01	lb/hr	0	0	0
P-XYL-01	lb/hr	0	0	0
1:3:5-01	lb/hr	0	0	0
N-PRO-01	lb/hr	0	0	0.00013
PHENA-01	lb/hr	0	0	0
CHRY-01	lb/hr	0	0	0
DIAMA-01	lb/hr	0	0	0
DIPHE-01	lb/hr	0	0	0
DECAH-01	lb/hr	0.00E+00	0	10.9
N-BUT-03	lb/hr	0	0	0
N-PRO-02	lb/hr	0	0	0
1:3:5-02	lb/hr	0	0	0
PROPA-01	lb/hr	0	0	0

ISOBU-01	lb/hr	0	0	0
1:1-D-01	lb/hr	0	0	0
N-TRI-01	lb/hr	0	0	0
BICYC-01	lb/hr	0	0	0
TETRA-01	lb/hr	0	0	0

Column1	Column2	201	202	203	204
Temperature	F	100	248	212	216.8
Pressure	psi	300	30.89	13.69	1469.5
H2	lb/hr	9541.98	9541.98	0	0
CH4	lb/hr	0	0	0	0
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	0	0	0	0
ACETONE	lb/hr	0	0	2834.29	2834.29
H2O	lb/hr	0	0	67.66	67.6
AACID	lb/hr	0	0	32789.2	32789.2
FURFURAL	lb/hr	0	0	8598.7	8598.7
MCRESOL	lb/hr	0	0	11902.9	11902.9
GUAIACOL	lb/hr	0	0	81792.3	81792.3
LVG	lb/hr	1.36761E-09	1.36761E-09	13004.8	13004.8
C2H6	lb/hr	0	0	0	0
C10H22	lb/hr	0	0	0	0
2-PHE-01	lb/hr	0	0	0	0
N-HEP-01	lb/hr	0	0	0	0
P-XYL-01	lb/hr	0	0	0	0
1:3:5-01	lb/hr	0	0	0	0
N-PRO-01	lb/hr	0.000130	0.000130	0	0
PHENA-01	lb/hr	0	0	0	0
CHRYC-01	lb/hr	0	0	0	0
DIAMA-01	lb/hr	0	0	0	0
DIPHE-01	lb/hr	0	0	0	0
DECAH-01	lb/hr	1.10E+01	10.98	0	0
N-BUT-03	lb/hr	0	0	0	0
N-PRO-02	lb/hr	0	0	0	0
1:3:5-02	lb/hr	0	0	0	0
PROPA-01	lb/hr	0	0	0	0
ISOBU-01	lb/hr	0	0	0	0

1:1-D-01	lb/hr	0	0	0	0
N-TRI-01	lb/hr	0	0	0	0
BICYC-01	lb/hr	0	0	0	0
TETRA-01	lb/hr	0	0	0	0

Column1	Column2	205	206	207	208
Temperature	F	185.3	185	149	149
Pressure	psi	30.8	1450.3	1449.3	414.69
H2	lb/hr	9541.9	3820.1	3820.1	3802.4
CH4	lb/hr	0	3332.8	3332.87	3271.8
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	0	8953.8	8953.8	8382.8
ACETONE	lb/hr	2834.2	0	0	0
H2O	lb/hr	67.6	71566.8	71566.8	399.74
AACID	lb/hr	32789.2	0	0	0
FURFURAL	lb/hr	8598.7	0	0	0
MCRESOL	lb/hr	11902.9	0	0	0
GUAIACOL	lb/hr	81792.3	0	0	0
LVG	lb/hr	13004.8	0.0010	0.00101	1.36761E-09
C2H6	lb/hr	0	1996.6	1996.6	1836.12
C10H22	lb/hr	0	1984.4	1984.4	33.5
2-PHE-01	lb/hr	0	7958.9	7958.9	11.5
N-HEP-01	lb/hr	0	3258.8	3258.8	272.6
P-XYL-01	lb/hr	0	1729.2	1729.2	34.50
1:3:5-01	lb/hr	0	638.8	638.8	4.78
N-PRO-01	lb/hr	0.00013	0.0130	0.013	0.00013
PHENA-01	lb/hr	0	6056.6	6056.6	0.0782
CHRYC-01	lb/hr	0	6088.9	6088.9	0.00030
DIAMA-01	lb/hr	0	8590.9	8590.9	12.53
DIPHE-01	lb/hr	0	4360.8	4360.8	1.31
DECAH-01	lb/hr	1.10E+01	3170.0	3170.08	10.98
N-BUT-03	lb/hr	0	232.9	232.9	1.71
N-PRO-02	lb/hr	0	9735.3	9735.3	144.21
1:3:5-02	lb/hr	0	3170.0	3170.	3.58
PROPA-01	lb/hr	0	1696.2	1696.2	1359.34
ISOBUT-01	lb/hr	0	1496.7	1496.7	995.3
1:1-D-01	lb/hr	0	1585.0	1585.0	13.16
N-TRI-01	lb/hr	0	8590.9	8590.9	6.39

BICYC-01	lb/hr	0	237.7	237.7	0.148
TETRA-01	lb/hr	0	289.9	289.9	0.004

Column1	Column2	209	S-B-OIL	WWATER1	210
Temperature	F	149	149	149	200
Pressure	psi	414.69	414.69	414.69	49.69
H2	lb/hr	760.49	17.47	0.176	16.80
CH4	lb/hr	3271.88	60.64	0.358	53.50
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	8382.88	562.76	8.17	398.07
ACETONE	lb/hr	0	0	0	0
H2O	lb/hr	399.74	628.40	70538.5	78.27
AACID	lb/hr	0	0	0	0
FURFURAL	lb/hr	0	0	0	0
MCRESOL	lb/hr	0	0	0	0
GUAIACOL	lb/hr	0	0	0	0
LVG	lb/hr	0	0.00091	9.15805E-05	1.17544E-09
C2H6	lb/hr	1836.12	160.2	0.27	103.57
C10H22	lb/hr	33.55	1950.9	1.35443E-11	9.29
2-PHE-01	lb/hr	11.53	7927.03	20.32	4.79
N-HEP-01	lb/hr	272.6	2986.18	3.89652E-07	64.76
P-XYL-01	lb/hr	34.5	1694.6	0.000180	9.792
1:3:5-01	lb/hr	4.789	634.0	1.69828E-06	1.517187772
N-PRO-01	lb/hr	0	0.012	4.66E-11	3.90881E-05
PHENA-01	lb/hr	0.078	6056.5	2.00292E-06	0.042
CHRYC-01	lb/hr	0.00030	6088.	4.917E-13	0.00
DIAMA-01	lb/hr	12.531	8578.4	8.23947E-16	3.385
DIPHE-01	lb/hr	1.3172	4359.5	3.04595E-05	0.553
DECAH-01	lb/hr	0.00E+00	3159.0	1.6749E-06	3.450
N-BUT-03	lb/hr	1.710	231.2	2.28112E-11	0.478
N-PRO-02	lb/hr	144.21	9591.0	9.7275E-08	38.66
1:3:5-02	lb/hr	3.587	3166.4	1.20631E-11	1.3373
PROPA-01	lb/hr	1359.3	336.9	0.039	138.73
ISOBUT-01	lb/hr	995.31	501.4	0.0019	134.48
1:1-D-01	lb/hr	13.16	1571.8	1.22296E-06	3.999
N-TRI-01	lb/hr	6.39	8584.5	1.92331E-19	2.419
BICYC-01	lb/hr	0.148	237.6	2.25016E-11	0.0503

TETRA-01	lb/hr	0.0049	289.93	7.21037E-47	0.00221
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Column1	Column2	211	WWATER 2	212	213
Temperature	F	200	206.7670766	206.76	207.99
Pressure	psi	49.6	49.69594878	49.69	214.69
H2	lb/hr	0.66	0	0.66	0.66
CH4	lb/hr	7.13	0	7.13	7.13
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	164.69	0	164.69	164.6
ACETONE	lb/hr	0	0	0	0
H2O	lb/hr	550.12	539.8875425	10.23	10.2
AACID	lb/hr	0	0	0	0
FURFURAL	lb/hr	0	0	0	0
MCRESOL	lb/hr	0	0	0	0
GUAIACOL	lb/hr	0	0	0	0
LVG	lb/hr	0.00091	0	0.00091	0.00091
C2H6	lb/hr	56.709	0	56.71	56.70
C10H22	lb/hr	1941.61	0	1941.61	1941.61
2-PHE-01	lb/hr	7922.2	0	7922.24	7922.24
N-HEP-01	lb/hr	2921.42	0	2921.42	2921.42
P-XYL-01	lb/hr	1684.90	0	1684.90	1684.90
1:3:5-01	lb/hr	632.49	0	632.49	632.4
N-PRO-01	lb/hr	0.012	0	0.01286	0.0128
PHENA-01	lb/hr	6056.5	0	6056.51	6056.5
CHRY-01	lb/hr	6088.9	0	6088.90	6088.9
DIAMA-01	lb/hr	8575.0	0	8575.04	8575.04
DIPHE-01	lb/hr	4358.94	0	4358.94	4358.94
DECAH-01	lb/hr	3.16E+03	0	3155.64	3155.64
N-BUT-03	lb/hr	230.758	0	230.758	230.75
N-PRO-02	lb/hr	9552.42	0	9552.42	9552.4
1:3:5-02	lb/hr	3165.15	0	3165.150	3165.15
PROPA-01	lb/hr	198.176	0	198.176	198.176
ISOB-01	lb/hr	366.93	0	366.93	366.936
1:1-D-01	lb/hr	1567.87	0	1567.87	1567.87
N-TRI-01	lb/hr	8582.14	0	8582.14	8582.140
BICYC-01	lb/hr	237.56	0	237.56	237.565
TETRA-01	lb/hr	289.93	0	289.93	289.932

Column1	Column2	FLUEGAS	S-OIL1	S-OIL2	DIESEL
Temperature	F	201.79	656.09	656.98	885.10
Pressure	psi	214.69	216.39	264.69	268.0
H2	lb/hr	0.66	9.95209E-12	9.95209E-12	5.95024E-31
CH4	lb/hr	7.13	3.11253E-09	3.11253E-09	6.78771E-27
C2H4	lb/hr	0	0	0	0
CO2	lb/hr	164.69	1.61972E-06	1.61972E-06	1.27643E-22
ACETONE	lb/hr	0	0	0	0
H2O	lb/hr	10.22	0.0075	0.0075	1.18785E-12
AACID	lb/hr	0	0	0	0
FURFURAL	lb/hr	0	0	0	0
MCRESOL	lb/hr	0	0	0	0
GUAIACOL	lb/hr	0	0	0	0
LVG	lb/hr	3.30434E-25	0.00091	0.00091	0.0005
C2H6	lb/hr	56.70	5.64116E-06	5.64116E-06	8.95106E-21
C10H22	lb/hr	0.00126	1941.61	1941.6	3.36
2-PHE-01	lb/hr	1.7433E-05	7922.24	7922.2	245.54
N-HEP-01	lb/hr	99.35	2822.07	2822.07	0.0062
P-XYL-01	lb/hr	0.0312	1684.87	1684.87	1.35
1:3:5-01	lb/hr	8.97189E-05	632.49	632.49	4.27
N-PRO-01	lb/hr	4.67906E-09	0.0128	0.0128	8.85919E-05
PHENA-01	lb/hr	6.93952E-17	6056.51	6056.51	3894.5
CHRY-01	lb/hr	1.69731E-28	6088.90	6088.90	4188.1
DIAMA-01	lb/hr	2.85876E-10	8575.04	8575.04	4955.6
DIPHE-01	lb/hr	2.00927E-10	4358.94	4358.94	2224.00
DECAH-01	lb/hr	2.66E-06	3155.64	3155.64	599.8
N-BUT-03	lb/hr	1.16943E-06	230.75	230.75	23.7
N-PRO-02	lb/hr	0.0032	9552.42	9552.42	152.8
1:3:5-02	lb/hr	1.34191E-08	3165.15	3165.15	698.5
PROPA-01	lb/hr	198.17	0.00094	0.00094	1.72158E-16
ISOB-01	lb/hr	366.9074144	0.02941	0.0294	1.62269E-13

1:1-D-01	lb/hr	0.000180466	1567.87	1567.87	19.5
N-TRI-01	lb/hr	4.7522E-10	8582.14	8582.14	2523.4
BICYC-01	lb/hr	2.81807E-11	237.56	237.56	110.03
TETRA-01	lb/hr	5.08837E-23	289.93	289.93	184.163

Column1	Column2	NAPTHA	RP1	RP-1
Temperature	F	573.4	705.02	705.02
Pressure	psi	264.6	293.91	293.91
H2	lb/hr	9.95209E-12	2.09284E-19	2.09284E-19
CH4	lb/hr	3.11253E-09	3.80721E-16	3.80721E-16
C2H4	lb/hr	0	0	0
CO2	lb/hr	1.61972E-06	1.14523E-12	1.14523E-12
ACETONE	lb/hr	0	0	0
H2O	lb/hr	0.0075	4.75578E-06	4.75578E-06
AACID	lb/hr	0	0	0
FURFURAL	lb/hr	0	0	0
MCRESOL	lb/hr	0	0	0
GUAIACOL	lb/hr	0	0	0
LVG	lb/hr	5.13834E-10	0.00033	0.00033
C2H6	lb/hr	5.64115E-06	1.5567E-11	1.5567E-11
C10H22	lb/hr	1106.5	831.68	831.68
2-PHE-01	lb/hr	2312.1	5364.56	5364.56
N-HEP-01	lb/hr	2693.	129.002	129.002
P-XYL-01	lb/hr	1078.7	604.81	604.81
1:3:5-01	lb/hr	277.2	350.96	350.96
N-PRO-01	lb/hr	0.0057	0.007065	0.00706
PHENA-01	lb/hr	0.0023	2161.99	2161.9
CHRY-01	lb/hr	1.59702E-10	1900.778	1900.77
DIAMA-01	lb/hr	5.5	3613.82	3613.82
DIPHE-01	lb/hr	39.6	2095.29	2095.29
DECAH-01	lb/hr	5.09E+02	2047.16	2047.16
N-BUT-03	lb/hr	51.7	155.31	155.31
N-PRO-02	lb/hr	3677.6	5721.97	5721.97
1:3:5-02	lb/hr	385.8	2080.65	2080.65
PROPA-01	lb/hr	0.00094	2.41866E-8	2.41866E-08
ISOB-01	lb/hr	0.029	3.69862E-06	3.69862E-06

1:1-D-01	lb/hr	614.92	933.37	933.37
N-TRI-01	lb/hr	616.07	5442.64	5442.6
BICYC-01	lb/hr	4.97	122.54	122.6
TETRA-01	lb/hr	2.50035E-05	105.76	105.76

3.8 Equipment List & Sizing

Table 5: Fast Pyrolysis Equipment List

Fast Pyrolysis			
Label	Equipment	Material	Sizing
D-100	Dryer	Carbon Steel	Tray Area = 200.21 sq ft
R-100	Pyrolysis rise reactor	316 SS	Diameter = 0.98 in Length = 20.01 ft
E-105	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft
E-106	HEX	316 SS	Diameter = 0.98 in Length = 20.01 ft
E-107	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft
R-101	Combustor	316 SS	Volume= 38.2 gal Diameter = 12.01 in Height = 6.5 ft
D-100	Quench Absorber	Carbon Steel	Volume= 4,894.1 gal Diameter = 6.99 ft Height = 17.06 ft

Table 6: Bio-Oil Upgrading Equipment List

Bio-Oil Upgrading			
Label	Equipment	Material	Sizing
R-200	Hydrotreater	1 1/4 Chrome	Volume = 2,022.55 gal Diameter = 4.49 ft Height = 17.06 ft
E-201	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft
E-202	Heater	316 SS	Diameter = 0.98 in Length = 20.01 ft
D-200A	Flash Drum	Killed Carbon Steel	Volume = 2,023.21 gal Diameter = 4.99 ft Height = 14.99 ft
D-200B	Flash Drum	Killed Carbon Steel	Volume = 2,023.21 gal Diameter = 4.99 ft Height = 14.99 ft
PSA	PSA	316 Stainless Steel	Volume = 634.52 gal Diameter = 2.99 ft Height = 12.01 ft

Table 7: RP-1 Separation Equipment List

RP-1 Separation			
Label	Equipment	Material	Sizing
C-300	Distilled Columns	Killed Carbon Steel	Diameter = 4 ft Height = 121.39
C-301	Distilled Columns	Killed Carbon Steel	Diameter = 4 ft Height = 121.39

3.9 Hand Calculations

To begin the process of sizing the debutanizer distillation column, the Fenske Equation is used to find $N_{\min}$:

$$N_{min} = \frac{\log \left[\left(\frac{x_{LK,D}}{x_{HK,D}} \right) \left(\frac{x_{HK,B}}{x_{LK,B}} \right) \right]}{\log (\alpha_{LK,HK})}$$

Also, the geometric mean of the relative volatility is used:

$$\alpha_{LK,HK} = \sqrt{(\alpha_D)(\alpha_B)}$$

To start with, the relative volatility of the distillate is found. Antoine's equation is used to obtain the saturated pressures of the light and heavy key.

$$\log(P^{sat}) = A - \frac{B}{T+C}$$

A, B, and C are Antoine's coefficients, obtained from NIST. The relationship between the saturated pressures of the light and heavy key is as follows:

$$\alpha_D = \frac{P_{LK}^{sat}}{P_{HK}^{sat}}$$

For the light key, A = 4.3281, B = 1132.108, C = 0.918, and T = 366.3 K.

$$P_{LK}^{sat} = 17.64 \text{ bar}$$

For the heavy key, A = 4.01356, B = 1461.715, C = -65.16, and T = 619.7 K.

$$P_{HK}^{sat} = 0.145 \text{ bar}$$

From here:

$$\alpha_D = \frac{P_{LK}^{sat}}{P_{HK}^{sat}} = \frac{17.64}{0.145} = 121.45$$

Moving on the relative volatility of the bottoms. Since the temperature of the bottoms is above the critical temperature of the light key, Antoine's equation is not valid. Instead, Wilson's correlation is used.

$$\ln(k) = \ln\left(\frac{P_c}{P}\right) + 5.373(1+w)\left(1 - \frac{T_c}{T}\right)$$

$$\alpha_B = \frac{k_{LK}}{k_{HK}}$$

For the light key: $P_c = 36.4$ bar, $w = 0.183$, and $T_c = 407.85$ K.

$$k_{LK} = 17.54$$

For the heavy key, $P_c = 28$ bar, $w = 0.258$, and $T_c = 639$ K.

$$k_{HK} = 1.245$$

$$\alpha_B = \frac{k_{LK}}{k_{HK}} = \frac{17.54}{1.245} = 14.09$$

From here, all values required for N_{min} are known. Plugging everything into the Fenske equation results in:

$$N_{min} = 6.54 = 7 \text{ stages}$$

Next, N and N_{actual} are found. N is found using the 1st and 2nd Underwood Equations and the Gilliland Correlation:

$$F(1-q) = \sum_{i=1}^C \frac{\alpha_{i-ref} F z_{F,i}}{\alpha_{i-ref} - \theta}$$

$$V_{min} = \sum_{i=1}^C \frac{\alpha_{i-ref} (D x_i)_D}{\alpha_{i-ref} - \theta}$$

$$\frac{V_{min}}{D} = R_{min} + 1$$

$$X = \frac{R - R_{min}}{R + 1}$$

$$Y = 1 - \exp\left[\frac{-1.33 X}{1 - X}\right] = \frac{N - N_{min}}{N + 1}$$

Plugging everything in grants values of:

$$\theta = 39.87$$

$$R_{min} = 26.7$$

$$X = 0.224$$

$$Y = 0.32$$

$$N = 10.3 \text{ stages}$$

To get N_{actual} , efficiency is taken into consideration. An assumption of 60% is used:

$$N_{actual} = \frac{N}{\eta}$$

$$N_{actual} = 17.2 = 18 \text{ stages}$$

Next, the column diameter and height are found. To get the column diameter, the Sauders Brown equation is used:

$$\text{Sauders Brown: } u_f = C \sqrt{\frac{\rho_L - \rho_V}{\rho_V}}$$

$$\text{Tray Capacity Constant}, C = 0.11 \frac{m}{s}$$

$$u_{des} (@ 80 \% \text{ flooding}) = 0.8 u_f$$

Next, everything is plugged into the diameter relationship:

$$D = \sqrt{\frac{4 \dot{V}}{\pi \phi u_{des}}}$$

$$\phi (\text{active area fraction}) = 0.8$$

$$D = 0.8 m = 2.6 ft$$

Next, to find the height of the column, the following equation is used:

$$H_{active} = (N_{trays} - 1) S$$

$$S (\text{tray spacing}) = 24 \text{ in} = 0.61 m = 2 ft$$

$$H_{active} = 10.4 m = 34 ft$$

Add an additional 3 m for tops and bottoms equipment

$$H = 13.4 m = 43 ft$$

4. Preliminary Utilities

Table 7: Preliminary Utility Summary

Preliminary Utilities Summary		
Type	Cost	Rate
Low Pressure Steam (\$/kWh)	\$ 0.022	2000 kWh
Cooling Water (\$/gal)	\$ 0.002	45,000 gal/hr
Electricity (\$/kWh)	\$ 0.06	1064.79 kWh
Wastewater (\$/gal)	\$ 0.002	10,097 gal/hr
Fired Heat (\$/kWh)	\$ 0.02	22,567 kWh

The various equipment in the process utilizes these utilities. For electricity, equipment such as pumps and compressors are the ones that both use and generate the overall electricity rate. Moreover, cooling water is used repeatedly throughout the process in various exchangers, condensers, etc. The use of cooling water is widespread and flexible, as it can cool a stream up to 25 °C. Low-pressure steam was also used to heat streams in heaters, exchangers, reboilers, and other components in columns. The same for fired heat, which is used for temperatures that steam cannot handle, such as reboilers in the de-butanization and fractionation of stable oil. The reaction of bio-oil in the hydrotreater yields immense amounts of wastewater with close to 99% water purity. But because of these impurities, it must be treated and sent to wastewater treatment. At a cost of \$ 2 per 1,000 gallons and about 45,000 gallons/hr of wastewater produced in the process, a hefty cost will be added to the wastewater. The pricing of these utilities was obtained from Dr. C, the EIA, and Mr. Bahr as sources. All in all, the utilities listed in the chart above are preliminary and subject to change, as high-pressure steam can be generated rapidly during fast pyrolysis. As well as other utilities, they can be replaced with fuel gas from the de-butanizer tops and a possible source of electricity.

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